in collaboration with

ST6003CEM Web API Development Project proposal

Doctor On Call

doctoroncall

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Project Introduction

Doctor On Call is an innovative web-based doctor appointment system designed to bridge the communication gap between patients and healthcare providers by leveraging modern web technologies. The platform provides real-time visibility into doctor availability and appointment status while streamlining booking and scheduling procedures through the use of the MERN stack (MongoDB, Express, React, and Node.js). Wheelchair care, nutrition, and cardiology are just a few of the categories that patients can easily use to find doctors, making it easier for them to get the right medical care.

This project integrates authentication and role-based access for patients, doctors, and administrators. Patients can create profiles, book and manage appointments, and receive notifications about their bookings. Doctors maintain their schedules and update availability, while administrators oversee the system, monitor appointments, and manage user data to maintain smooth operations within the healthcare environment.

Justification

Healthcare facilities frequently face challenges with manual or fragmented appointment systems, often resulting in long waiting periods, scheduling conflicts, and patient dissatisfaction. This situation not only frustrates patients but also puts a strain on administrative personnel, who must handle numerous calls and walk-in requests to manage appointments. Present-day technologies offer tremendous opportunities to modernise these processes through digital platforms.

The real-world need for a comprehensive system that provides patients and healthcare providers with clear, transparent, and effective appointment management served as the impetus for Doctor On Call. Digital transformation in healthcare is not just a convenience but a necessity to improve service delivery, reduce redundant workloads for staff, and enhance the overall patient experience. Doctor On Call seeks to decrease waiting times, minimise appointment overlap, and lower no-show rates through timely notifications by combining appointment scheduling and communication into a centralised, automated platform.

This platform addresses a critical gap, especially in high-traffic clinics, by offering real-time updates on queue status and doctor availability, enabling patients to plan visits better and avoid unnecessary waiting. This strategy is in line with current healthcare trends that prioritise operational effectiveness and patient-centric services.

Objective of the Project

The primary objective of Doctor On Call is to create a fully functional, scalable web application that simplifies appointment booking and management within healthcare settings. Particular objectives consist of:

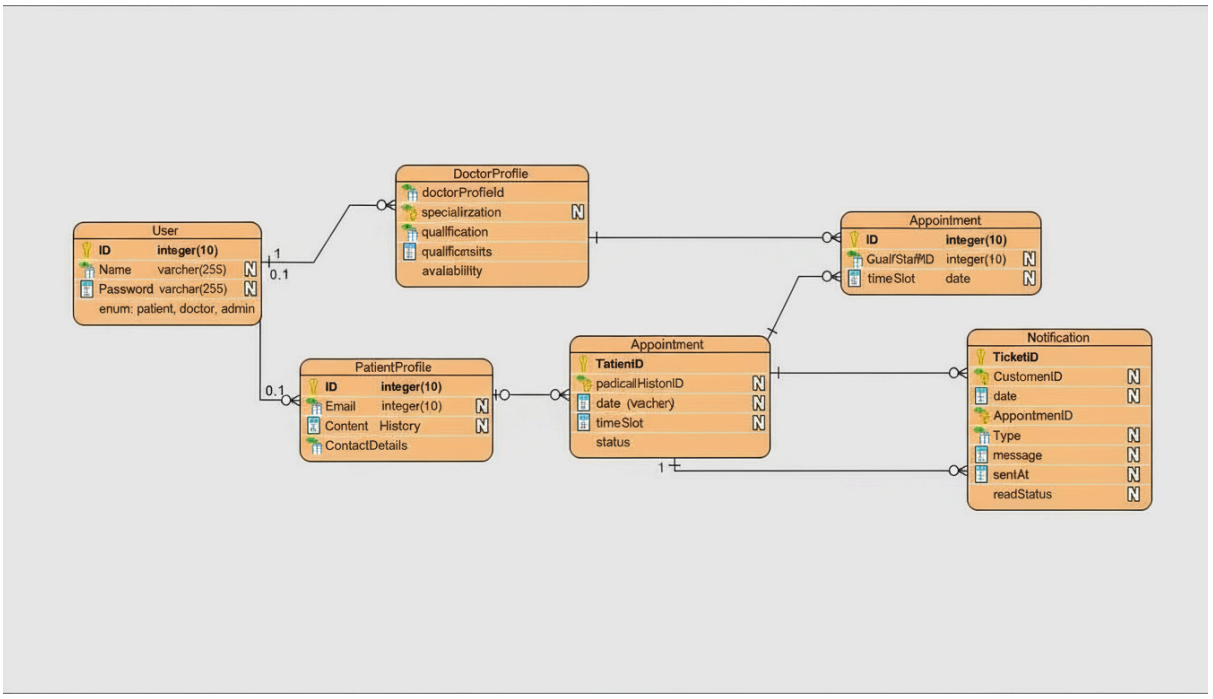
- Developing a responsive web interface enabling patients to search for doctors, view profiles, and book appointments effortlessly.
- Implementing a robust backend API using Node.js and Express.js to handle appointment logic, user authentication, and data management securely.
- Designing a flexible data model within MongoDB to store users, doctors, appointments, and notifications with scalability in mind.
- Ensuring secure user authentication and role-based access control via JWT to protect sensitive information.
- Enabling doctors to efficiently manage their availability and review upcoming appointments through dedicated dashboards.
- Providing administrators with comprehensive tools to monitor bookings, manage users, and oversee the operational health of the system.
- Implementing a notification system that sends automated reminders and updates to reduce missed appointments and enhance communication.

ERD Diagram

An Entity-Relationship Diagram for Doctor On Call includes the following key entities and relationships:

- User (role: patient, doctor, admin) with attributes: userId, name, email, password, role.
- DoctorProfile related to User where role = doctor; attributes include specialization, qualifications, availability.
- PatientProfile linked to User where role = patient; includes medical history and contact details.
- Appointment with attributes: appointmentId, patientId, doctorId, date, timeSlot, status.
- Notification linked to appointments for reminders and updates.

This ERD supports flexible relation mapping for appointments between patients and doctors while maintaining user roles for system access control.



After Project Completion: Project Mission

Doctor On Call aims to transform outpatient healthcare experiences by offering a trustworthy and easy-to-use digital platform. The mission is to empower patients to access healthcare services with minimal friction while enabling medical professionals to focus more on patient care rather than administrative duties. By automating bookings and improving communication, the system will support operational efficiencies and contribute positively to patient satisfaction and healthcare outcomes.

Scope of Work

This project will be developed entirely using the MERN stack, offering a cohesive and high-performance environment for full-stack development:

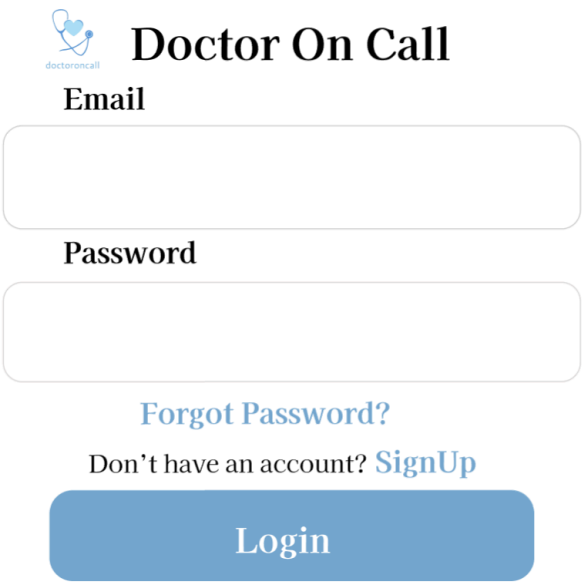
- MongoDB: A NoSQL document-oriented database will house user profiles, appointment records, and notification logs. It offers scalability and flexibility to adapt as user demand grows.
- Express.js: The backend framework will manage API endpoints to create, read, update, and delete appointment records securely and efficiently.
- React.js: The chosen frontend library will deliver a performant single-page application with dynamic UI updates reflecting real-time appointment slots and queue status.
- Node.js: A JavaScript runtime environment enabling asynchronous, event-driven backend processing that supports real-time data operations.
- JWT (JSON Web Tokens): Employed for stateless, secure user authentication across different user roles.
- Figma: Used as a design tool for prototyping the user interface to ensure consistent UX and UI design between web and mobile platforms.
- Additional Tools: Postman for API testing, GitHub for version control, and deployment platforms like Heroku or Netlify for hosting.

Features of the Project

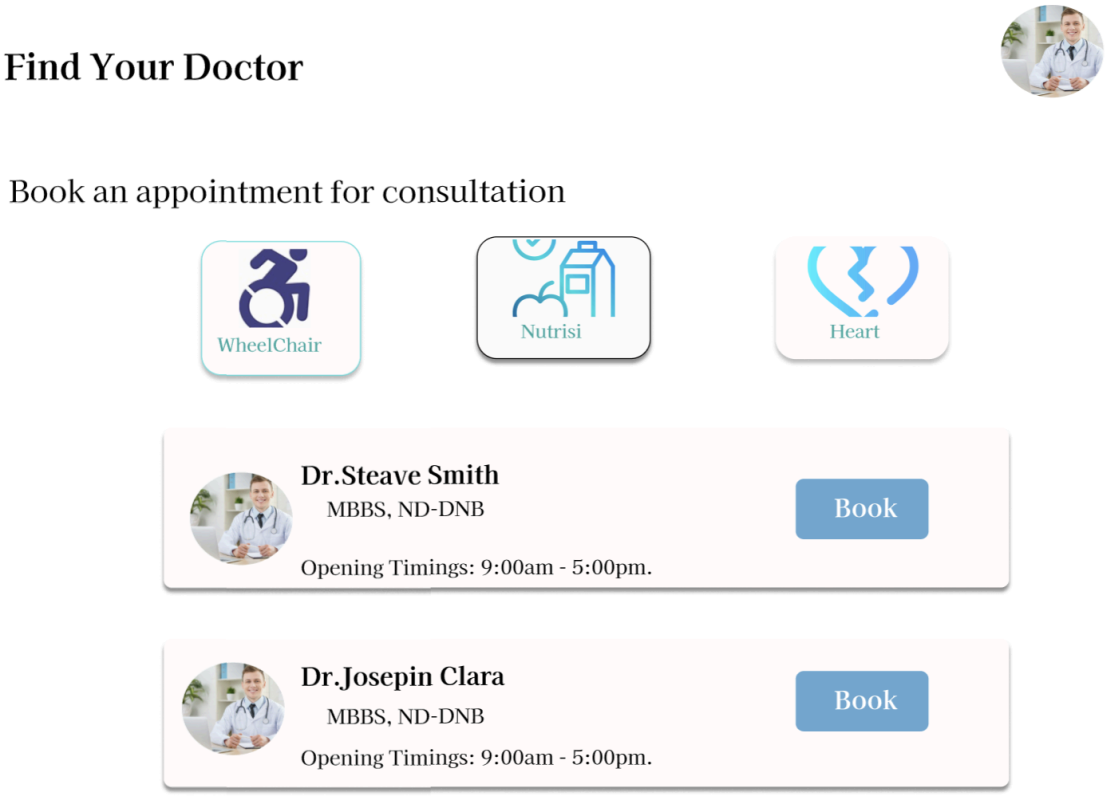
- User Registration and Login: Secure signup and login based on role with validation and password encryption.
- Doctor Browsing and Profile: Patients can browse doctors by categories such as Wheelchair, Nutrition, Heart care etc., view doctor details and qualifications.
- Appointment Booking: Patients can check available time slots (e.g., 9:00am - 5:00pm), select slots, and book appointments with instant confirmation.
- Appointment Management: Options to reschedule or cancel appointments with status updates reflected in real-time.
- Role-Based Dashboards: Separate dashboards for patients (view bookings), doctors (manage schedules), and admins (overview and controls).
- Notification System: Automated notifications and reminders for upcoming appointments via email or in-app alerts.
- Admin Controls: Manage users, doctors' schedules, oversee bookings, and monitor system performance.
- Secure Data Handling: All sensitive data is encrypted and API endpoints are protected with authentication checks.

Prototypes

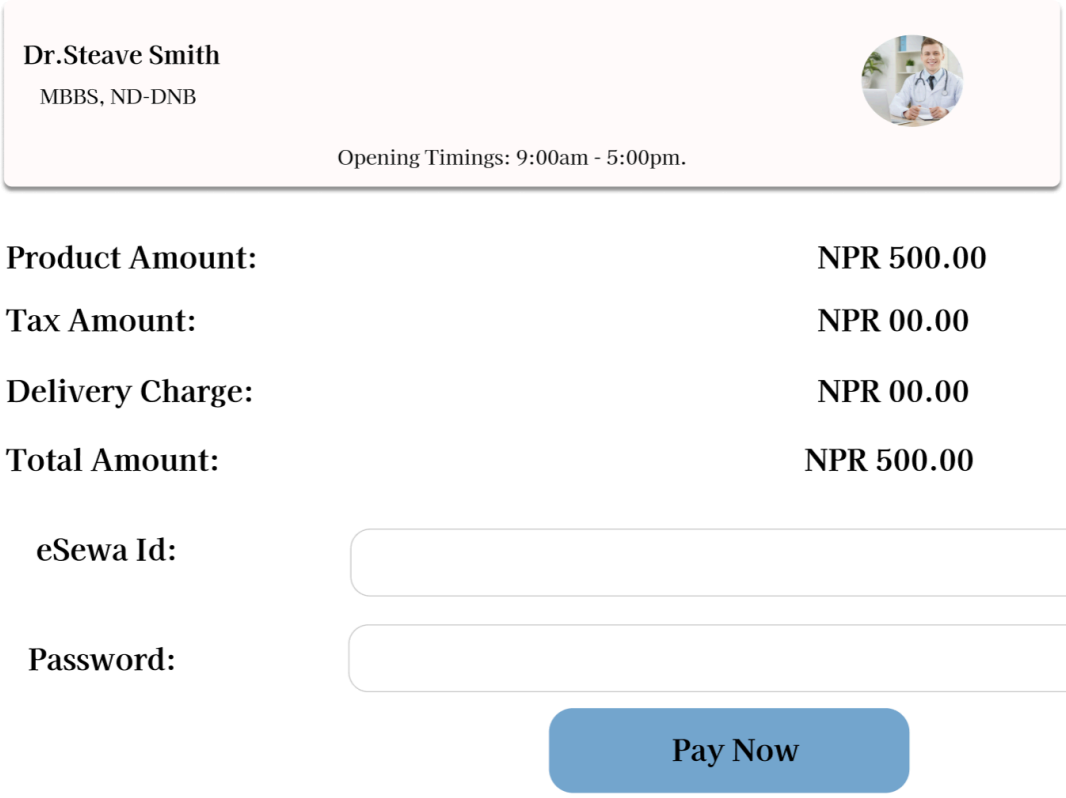
1. Login Page: Clean and simple login screen supporting email/password input with forgot password link.



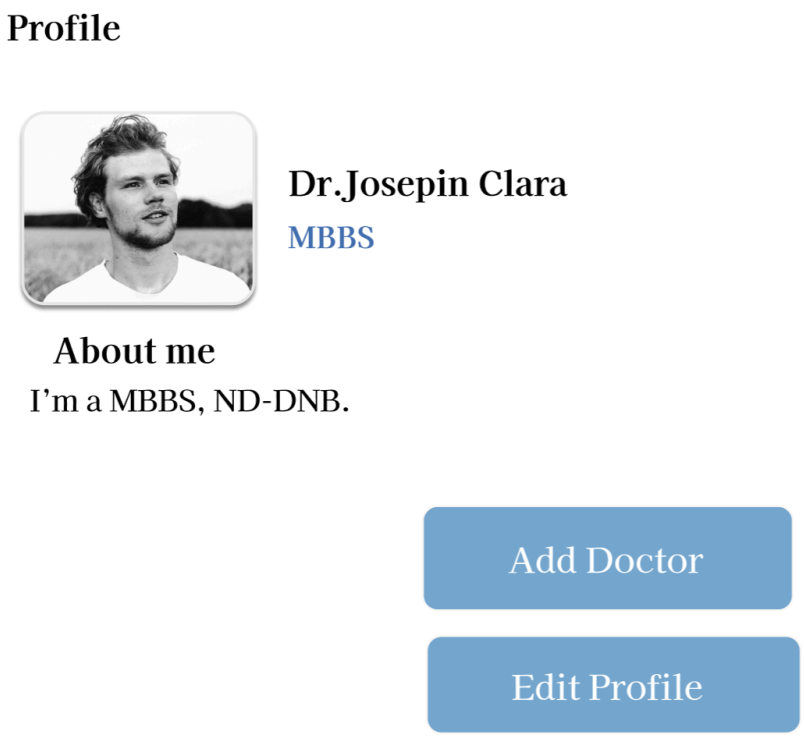
2. Dashboard: Main landing page featuring doctor categories, featured doctors, and shortcut buttons to book appointments.



3. Appointment Booking Page: Interactive calendar displaying selectable time slots, with options to confirm or update bookings.



4. Profile Page : To see own details and add new doctor.



Conclusion

Doctor On Call successfully addresses the common inefficiencies in healthcare appointment scheduling through a modern, web-based solution using the MERN stack. With features designed around ease of use and real-time updates, it enhances both patient experience and administrative workflow. The project's scalable design and secure framework make it a practical tool for clinics and medical institutions seeking to digitalize appointment management and improve service delivery.

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